Inverted MAML Card 1: Round 8 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.04, L2=1536, Mom=0.82, Clamp=±75 LSB

1. Round 8 Summary & Cumulative Comparison (R1–R8)

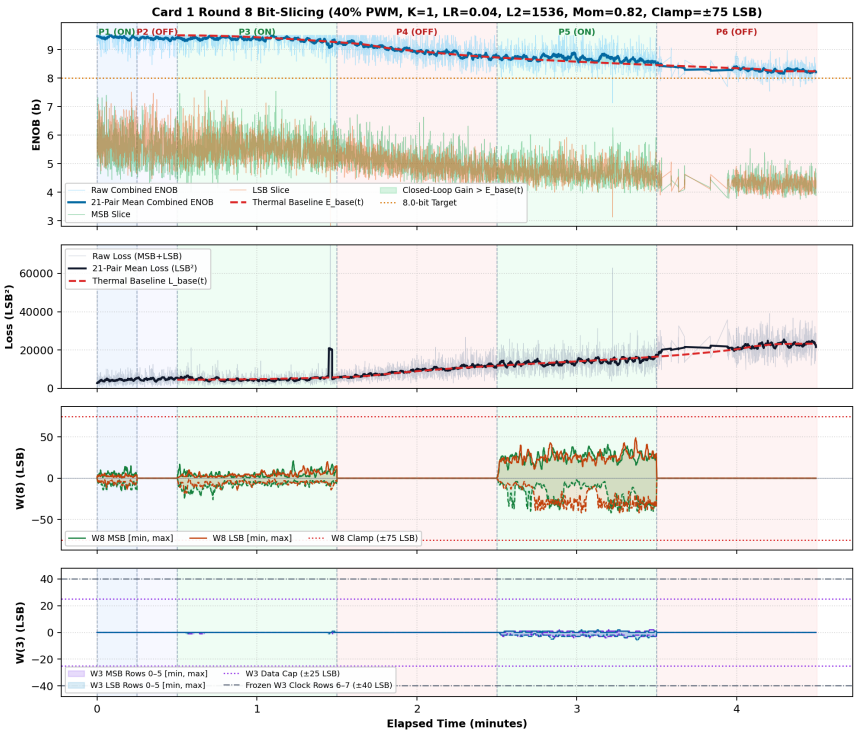
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 8 Detrended Trend Metrics:** Lift vs E_base = **+0.09 b**, Loss Ratio = **1.14x**, Climb = **+0.24 b**, Jitter = **0.26 b**, Recovery = **1.00**, Score = **+0.626**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1	0.03	1024	0.9	±75	+0.66b	1.57x	+0.74b	0.33b	0.67	+1.957
R2*	0.035	1536	0.85	±75	+0.50b	1.38x	+0.82b	0.29b	0.96	+2.004
R3	0.02	2560	0.8	±70	+0.09b	1.11x	+0.22b	0.26b	1.00	+0.559
R4	0.045	2048	0.88	±75	+0.11b	1.17x	+0.14b	0.24b	1.00	+0.643
R5	0.03	1024	0.85	±75	+0.20b	1.34x	+0.43b	0.26b	0.89	+1.147
R6	0.025	1280	0.92	±75	+0.06b	1.12x	+0.34b	0.23b	1.00	+0.689
R7	0.035	768	0.9	±75	+0.18b	1.38x	+0.48b	0.26b	0.95	+1.191
R8	0.04	1536	0.82	±75	+0.09b	1.14x	+0.24b	0.26b	1.00	+0.626

2. Phase-by-Phase Telemetry (3,931 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	342	9.46->9.40 (9.42b)	-	3049->4432	[16,14]	0	342
P2	Cold / Regler OFF	279	9.35->9.37 (9.39b)	-	5265->4988	[0,0]	0	0
P3	Warm / Regler ON	1061	9.43->9.32 (9.37b)	-0.01 b	4780->5784	[26,20]	1	1061
P4	Warm / Regler OFF	941	9.24->8.81 (8.97b)	+0.01 b	5806->11036	[0,0]	0	0
P5	Warm / Regler ON	838	8.74->8.66 (8.69b)	+0.16 b	12381->14639	[42,49]	6	838
P6	Warm / Regler OFF	470	8.43->8.26 (8.29b)	+0.02 b	18684->23236	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 8

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±75 LSB, while W(3) rows 0–5 stay within ±25 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 8 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

